

ESSI-FR v1: FRVT 1:1 local validation runs

These are three runs of NIST's official `run_validate_11.sh` with the ESSI-FR v1 recognition model. **Bold** marks a Run 3 value that differs from Run 2.

	Run 1	Run 2	Run 3
What changed	first clean run	same code and models, different machine	face detector swapped
Machine	VPS <code>shah-ho-vps</code>	AWS g5.xlarge	AWS g5.xlarge
Face detector	InsightFace SCRFD	InsightFace SCRFD	YuNet
Result	✔ PASS	✔ PASS	✔ PASS

1. Setup

	Run 1: VPS · SCRFD	Run 2: AWS · SCRFD	Run 3: AWS · YuNet
Date (IST)	18 Sep 2026, 23:23 – 23:59	19 Sep 2026, 11:55 – 12:05	19 Sep 2026, 12:34 – 12:38
CPU	QEMU virtual CPU, 10 vCPU	AMD EPYC 7R32, 4 vCPU	AMD EPYC 7R32, 4 vCPU
RAM	15 GB	15 GB	15 GB
OS the script ran on	Ubuntu 24.04.3 LTS (native)	Ubuntu 24.04.3 LTS (Docker <code>essi-frvt11:24.04.3</code> ; host is 26.04.1)	same as Run 2
Compiler / CMake	GCC 13.3.0 / 3.28.3	GCC 13.3.0 / 3.28.3	GCC 13.3.0 / 3.28.3
Compute	CPU, 1 thread	CPU, 1 thread	CPU, 1 thread
Detector file	<code>det_10g.onnx</code> (16.9 MB)	<code>det_10g.onnx</code> (16.9 MB)	<code>face_detection_yunet_2023mar.onnx</code> (232 KB)
Detector source / licence	InsightFace SCRFD-10G, non-commercial research	same	OpenCV Zoo YuNet, MIT
Detector SHA256	<code>5838f7fe...</code>	<code>5838f7fe...</code>	<code>8f2383e4...</code>
Detector input	RGB, (px – 127.5) / 128	same	BGR, raw 0–255
Detector scales	1 pass at 640	1 pass at 640	3 passes (640 / 320 / 160), merged
Score threshold / NMS	0.5 / 0.4	0.5 / 0.4	0.6 / 0.3

	Run 1: VPS · SCRFD	Run 2: AWS · SCRFD	Run 3: AWS · YuNet
Recognition model	<code>essi_fr_v1_last.onnx</code> (68726933...)	same	same
Alignment	5-point, 112×112	same	same
Runtime	ONNX Runtime 1.20.1	same	same
Library	<code>libfrvt_11_essi_000.so</code> (80,880 B)	same (80,880 B)	85,456 B
Source changes	—	none (only the ONNX Runtime path in <code>CMakeLists.txt</code>)	<code>face_engine.cpp/.h</code> detection rewritten; comment in <code>essiimplfrvt11.cpp</code>
Package config/	<code>det_10g.onnx</code> , <code>essi_fr_v1_last.onnx</code>	same	<code>face_detection_yunet_2023mar.onnx</code> , <code>essi_fr_v1_last.onnx</code>
Package doc/	<code>version.txt</code>	<code>version.txt</code>	<code>version.txt</code> , <code>THIRD_PARTY_NOTICES.txt</code>
Package size	186.3 MB	186.3 MB	170.8 MB

2. NIST checks

Check	Run 1	Run 2	Run 3
OS version (24.04.3)	✓	✓	✓
No hard-coded config path	✓	✓	✓
Single-threaded (no warning)	✓	✓	✓
Works with <code>fork()</code> (4 processes)	✓	✓	✓
Multi-image and multi-person templates	✓	✓	✓
≥ 50% unique match scores	✓ 614/653	✓ 614/653	✓ 651/653
No negative scores	✓	✓	✓
Exit code / package created	0 / ✓	0 / ✓	0 / ✓

3. Face detection

	Run 1	Run 2	Run 3
Enroll: no face found (of 665)	29	29	2
· NIST test images ¹	5	5	2 (white, black)
· ordinary close-up photos	24	24	0
Verif: no face found (of 653)	15	15	0
Multi-person: templates written	34	34	28
Multi-person: images with no face	5	5	2 (white, black)
Group photos <code>multi1</code> / <code>multi2</code> / <code>multi3</code> : faces found ²	2 / 5 / 2	2 / 5 / 2	1 / 3 / 2
Eye positions	—	identical to Run 1	median 3.5% of face width from Run 2

4. Matching

Scores run 0–100, where score = (cosine + 1) × 50.

	Run 1	Run 2	Run 3
Pairs usable (of 653)	614	614	653
· same person / different person	367 / 247	367 / 247	389 / 264
· with a failed template (score 50)	39	39	0
Same person: lowest / mean score	60.39 / 88.10	60.39 / 88.10	62.86 / 87.66
Different person: highest / mean score	65.58 / 51.18	65.58 / 51.18	65.08 / 51.22
Best accuracy (at score)	99.84% (66.0)	99.84% (66.0)	99.85% (66.4)
Same-person pairs below the highest different-person score	1	1	1
Largest score difference vs Run 1	—	0.0002	—
Template size	2,048 B (512 floats)	2,048 B	2,048 B

5. Time (min:sec)

Step	Run 1	Run 2	Run 3
Hard-coded config check (10 images)	0:22	0:05	0:03
Enroll, 1 process (665 images)	22:04	4:59	2:06
Enroll, multiple images per subject (8)	1:00	0:14	0:05
Enroll, 4 processes (665)	5:37	2:15	0:55
Verif, 4 processes (653)	5:35	2:13	0:55
Match, 4 processes (653)	0:01	0:01	0:00
Verif, multi-person	0:48	0:11	0:06
Match, multi-person	0:01	0:01	0:00
Create package	0:11	0:09	0:09
Total	35:45	10:13	4:23
Time per image, 1 process	1.99 s	0.45 s	0.19 s

6. Where to find each run

	Logs	Package
Run 1	VPS FRS_NIST/frvt_logs/frvt11_essi_fr_v1_20260918_232320.*	VPS FRS_NIST/frvt/11/libfrvt_11_essi_000.v3
Run 2	AWS ~/frvt_logs/frvt11_essi_fr_v1_20260919_062523.* and ~/frvt_logs/baseline_scrfd_20260919/	AWS ~/frvt_previous_run_backup_20260919_070
Run 3	AWS ~/frvt_logs/frvt11_essi_fr_v1_20260919_070424.*	AWS ~/frvt/11/libfrvt_11_essi_000.v3.1.

¹ NIST's deliberate test images: `white.ppm`, `black.ppm`, `8bit.pgm` (greyscale), `rotated180.ppm`, `small.ppm` (7×9 px). Only white and black contain no face. ² Distinct faces. NIST's input lists each group photo twice, so the log has two rows per face. YuNet skips people who are partly hidden in the background; in `multi1` that person scored 0.52, just under the 0.6 threshold.